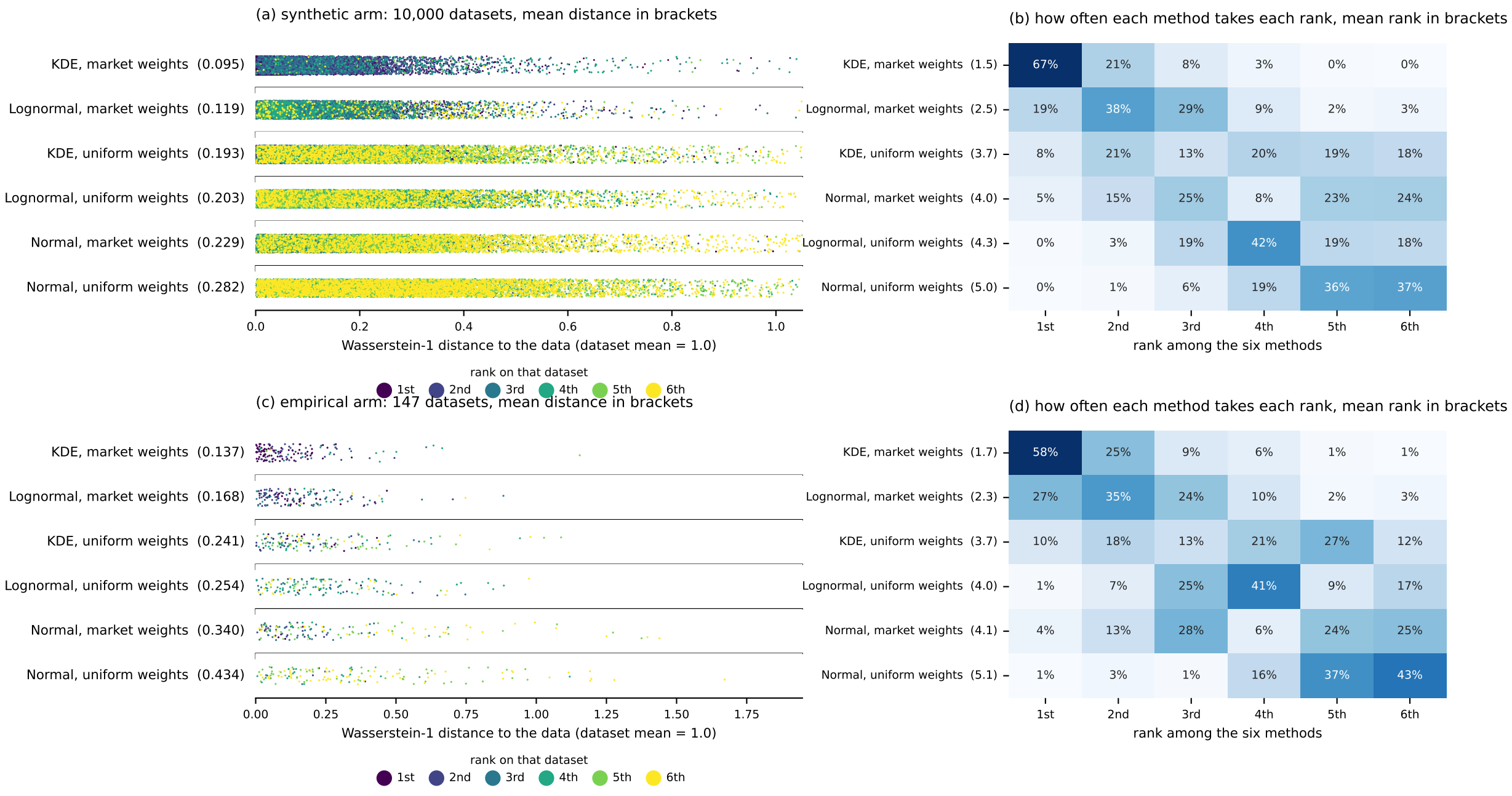

(a) synthetic arm: 10,000 datasets, mean distance in brackets
KDE, market weights (0.095)
Lognormal, market weights (0.119)
KDE, uniform weights (0.193)
Lognormal, uniform weights (0.203)
Normal, market weights (0.229)
Normal, uniform weights (0.282)
0.0
0.2
0.4
0.6
0.8
1.0
Wasserstein-1 distance to the data (dataset mean = 1.0)
rank on that dataset
1st
2nd
3rd
4th
5th
6th
(b) how often each method takes each rank, mean rank in brackets
KDE, market weights (1.5)
67%
21%
8%
3%
0%
0%
Lognormal, market weights (2.5)
19%
38%
29%
9%
2%
3%
KDE, uniform weights (3.7)
8%
21%
13%
20%
19%
18%
Normal, market weights (4.0)
5%
15%
25%
8%
23%
24%
Lognormal, uniform weights (4.3)
0%
3%
19%
42%
19%
18%
Normal, uniform weights (5.0)
0%
1%
6%
19%
36%
37%
1st
2nd
3rd
4th
5th
6th
rank among the six methods
(c) empirical arm: 147 datasets, mean distance in brackets
KDE, market weights (0.137)
Lognormal, market weights (0.168)
KDE, uniform weights (0.241)
Lognormal, uniform weights (0.254)
Normal, market weights (0.340)
Normal, uniform weights (0.434)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Wasserstein-1 distance to the data (dataset mean = 1.0)
rank on that dataset
1st
2nd
3rd
4th
5th
6th
(d) how often each method takes each rank, mean rank in brackets
KDE, market weights (1.7)
58%
25%
9%
6%
1%
1%
Lognormal, market weights (2.3)
27%
35%
24%
10%
2%
3%
KDE, uniform weights (3.7)
10%
18%
13%
21%
27%
12%
Lognormal, uniform weights (4.0)
1%
7%
25%
41%
9%
17%
Normal, market weights (4.1)
4%
13%
28%
6%
24%
25%
Normal, uniform weights (5.1)
1%
3%
1%
16%
37%
43%
1st
2nd
3rd
4th
5th
6th
rank among the six methods